

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Bit Banging Interface Definition
Applied to: USB4 Specification Versions 2.0

Brief description of the functional changes:

Adds more details regarding the TBT3-Compatibility bit banging mechanism in Chapter 13.

Benefits as a result of the changes:

Greater clarity in how bit banging is implemented.
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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None

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Actual Change

(a). Section 13.4.5 Bit Banging Interface

To Text:

This section only applies to a Device Router that supports TBT3-Compatibility on its Upstream Facing Port.

A Router shall support the “bit banging” interface defined in Vendor Specific 1 Capability (see Section 13.6.1.1).

A Router shall respond when the byte at address 0Ah is read. The Router may return any value.

A Router shall return the value 00000080h when the 32 bits at addresses [78h:75h] are read.

A Router shall return the value 01h when the byte at address 0148h is read.

A Router shall return the value 00000111h when the 32 bits at addresses [1A7h:1A4h] are read.



CONNECTION MANAGER NOTE

A Connection Manager reads the contents of a DROM attached to a TBT3 Router using the “bit banging” interface ~~defined in Vendor Specific 1 Capability (see Section 13.6.1.1).~~ The bit banging interface uses four bits (FL SK, FL CS, FL DI, and FL DO) to read from the flash memory of a device. The Connection Manager toggles the FL CS, FL CK, and FL DI bits as defined in DW 4 of the Vendor Specific 1 Capability (see Section 13.6.1.1). The Connection Manager reads the resulting data output from the FL DO bit.

When using the bit banging interface, a Connection Manager sends the following bytes on FL DI:

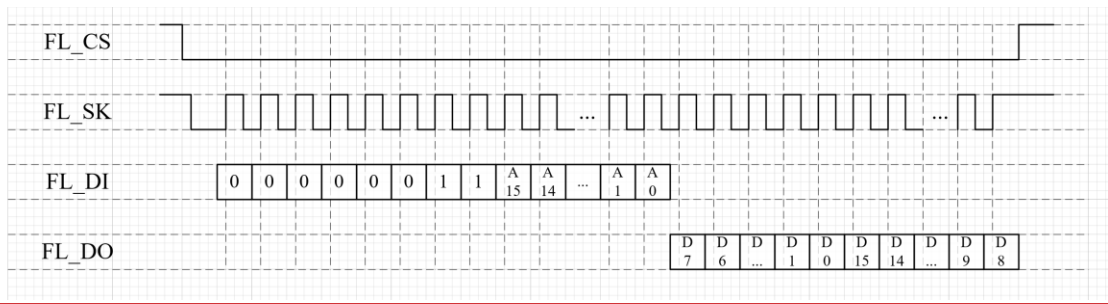
- Byte 1 is a read command (03h)*
- Byte 2 is bits [15:8] of the address to be read*
- Byte 3 is bits [7:0] of the address to be read*

The Connection Manager then reads the resulting data from the FL DO bit as follows:

- Byte 1 is bits [7:0] of the read data*
- Byte 2 is bits [15:8] of the read data (if present)*

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The waveform below shows an example of how the FL_SK, FL_CS, FL_DI, and FL_DO are used:



Note that all bytes are sent and received with the most significant bit first.

A Connection Manager can read from DROM or from any of the addresses listed in this section. When reading DROM, the base address can be read from the DROM Base Address field. The DROM Base Address field contains the DROM base address.

(b). Table 13-15 Vendor Specific 3 Capability Fields

To Text:

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
12	VSC_CS_12	31:0	DROM Base Address This field shall contain the base address (in bytes) of the DROM within the Flash Memory address space. This field shall be between 0000_0001h and (0000_FFFFh minus the DROM size).	RO	Vendor Defined
13 to 29		31:0	TBT3-Compatible	TBT 3	0